



slingshot college
(इस्लिङ्टन कलेज)

Module Code & Module Title

CC6057NI Applied Machine Learning

Individual Coursework

Submission: Milestone 1

Academic Semester: Autumn Semester 2025

Credit: 15 credit semester long module

Student Name: Resha Shrestha

London Met ID: 23048661

College ID: 230121

Assignment Due Date: 15/12/2025.

Assignment Submission Date: 15/12/2025

Submitted To: Mr. Mahotsav Bhattarai

GitHub Link	https://github.com/ReshaShrestha2004/Bloom-s-Taxonomy-Classification-of-Educational-Questions
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I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

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1. INTRODUCTION

1.1 Overview of the text classification project

Text classification is a key task in natural language processing (NLP) and machine learning, allowing automated systems to comprehend, sort, and react to human language.

The uses of text classification cover many real-world areas, such as spam detection, sentiment analysis, document categorization, recommendation systems, and educational technology. As educational platforms increasingly integrate artificial intelligence, the demand for smart processing of academic content has significantly increased.

Educational questions are a distinct and complex type of text data. Unlike casual text like social media posts, educational questions are crafted with specific teaching goals in mind. These goals are often implicit, conveyed through instructional verbs, sentence structure, and contextual hints. To automatically grasp these goals, one needs to perform not just syntactic analysis but also semantic interpretation.

Bloom's Taxonomy is a structured framework commonly used in education to categorize learning goals based on cognitive difficulty. The taxonomy arranges cognitive abilities into levels like Knowledge, Comprehension, Application, Analysis, Evaluation, and Creation. These levels assist educators in creating balanced assessments, assessing student learning results, and aligning teaching methods with learning goals. Even though it is significant, the classification of Bloom's Taxonomy is usually done manually. This method is labor-intensive, subjective, and can lead to inconsistencies, particularly when handling large amounts of questions. Consequently, there is a strong interest in exploring automated methods that can scale effectively while ensuring reasonable accuracy. This project tackles this issue by examining if machine learning models, trained on labeled educational question datasets, can successfully identify patterns linked to various Bloom's cognitive levels.

1.2 Project Scope and Objectives

The main goal of this project is to create and assess a text classification system using machine learning that can predict the cognitive level of an educational question based on Bloom's Taxonomy. The project highlights:

- Careful construction and analysis of the dataset
- Thoughtful pre-processing that maintains pedagogical meaning
- Explainable techniques for feature extraction
- Thorough evaluation methods

Instead of using complex deep learning models, this project deliberately concentrates on classical machine learning methods to guarantee interpretability, reproducibility, and a solid theoretical foundation which are qualities that are crucial for educational purposes.

1.3 Introduction to the machine learning and NLP concepts applied in this project.

Machine learning and natural language processing (NLP) offer computational techniques that allow systems to identify patterns in text data and make predictions without needing explicit programming rules. This project uses these methods to classify educational questions based on Bloom's Taxonomy cognitive levels.

1.3.1 Machine Learning Paradigm

This project is designed as a supervised multi-class classification task, where each educational question is linked to a specific cognitive level label.

Supervised Machine Learning involves using labeled datasets to train algorithms that accurately predict outcomes or classify data. The input dataset is fed into the supervised Machine Learning model, which adjusts its weights until the model is properly fitted, a process that is part of cross-validation. Supervised learning helps organizations tackle various real-world issues, it seeks algorithms that externally fit the instances to generate general hypotheses, enabling predictions for future cases (Rathindra Nath Mohalder, 2024).

A classification model is a kind of machine learning model that organizes data points into specific groups known as classes. Classifiers understand the features of each class from the input data, and then they learn to categorize new, unseen data based on those features (Belcic, n.d.).

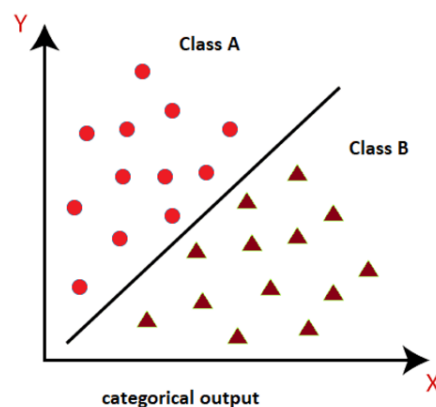


Figure 1 Classification (R, 2024)

Classification-based predictive modeling tasks are differentiated by the number of categories and how exclusive those categories are:

- Binary classification divides data into two exclusive categories.
- Multiclass classification divides data into more than two exclusive categories.
- Multilabel classification divides data into nonexclusive categories.
- Imbalanced classification features an unequal distribution of data points among categories (Belcic, n.d.).

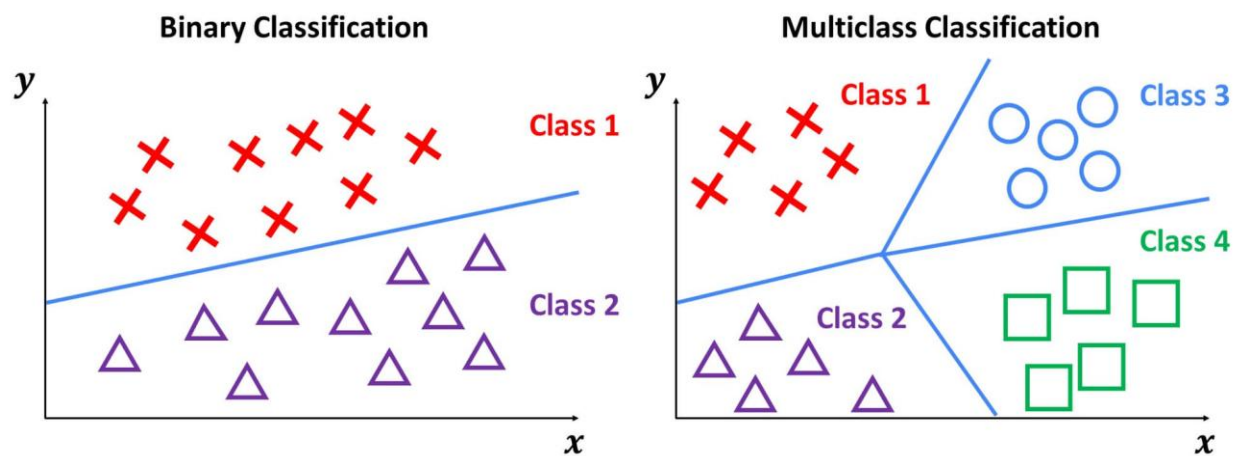


Figure 2 Types of classification (Zivkovic, 2022)

1.3.2 Text Cleaning and Pre-processing

Data gathered from real educational sources is often messy, inconsistent, and unstructured. Before any useful feature extraction or model training can take place, it is crucial to clean and standardize the raw text to maintain consistency while keeping the meaning of the educational questions intact. When it comes to classifying according to Bloom's Taxonomy, pre-processing choices need to be made very carefully. Identifying cognitive levels depends significantly on how instructions are phrased, the use of functional words, and verbs. If pre-processing is too aggressive, it might eliminate or alter these important language signals, resulting in a loss of teaching information. Thus, a careful yet minimal pre-processing approach was used in this project.

1.3.2.1 Lowercasing

All questions were converted to lowercase to eliminate case-sensitive inconsistencies in the vocabulary. Without this step, words such as *Define*, *define*, and *DEFINE* would be treated as distinct tokens, unnecessarily increasing vocabulary size which would ultimately lead to a larger dimension of word corpus.

Lowercasing improves:

- Vocabulary consistency
- Feature reliability
- Statistical robustness of term-based representations such as TF-IDF

Importantly, lowercasing does not alter semantic meaning in English educational text, making it a safe and effective normalization step.

1.3.2.2 Whitespace Normalization

Educational datasets obtained from spreadsheets and various sources frequently have inconsistent spacing caused by formatting issues. To fix this, leading and trailing whitespaces were eliminated, and several internal spaces were reduced to a single space.

Whitespace normalization is done in order to:

- Clear token boundaries for subsequent processing
- Avoidance of empty or incorrectly formatted tokens
- Consistent text representation throughout all samples

This process enhances textual uniformity without altering the order or meaning of the words.

1.3.2.3 Punctuation Elimination

Punctuation marks were taken out to lessen noise and dimensionality in the text representation. Although punctuation can provide stylistic details in certain NLP tasks, it does not play a major role in identifying Bloom's cognitive levels.

Eliminating punctuation:

- Makes tokenization easier
- Lowers feature sparsity
- Stops unnecessary changes in word forms

Numeric characters were kept intentionally, as numerical references might suggest application-oriented or problem-solving questions, which are important to Bloom's Taxonomy.

1.3.2.4 Preservation of Language Meaning

A key feature of the preprocessing approach in this project is the intentional omission of some frequently used text normalization methods:

- Stop word removal was not used because function words like how, why, what, and which often have a significant role in indicating cognitive intent.
- Stemming and lemmatization were not utilized to maintain verb tense and syntactic structure, which are vital signs of instructional depth.

This choice is consistent with educational NLP research that indicates keeping the natural language structure enhances performance in tasks related to intent and cognitive-level classification.

By steering clear of harsh changes, this preprocessing pipeline guarantees that:

- Instructional verbs stay unchanged
- Sentence structure is maintained
- Pedagogical meaning and linguistic richness are kept intact
- Irrelevant noise is minimized
- A clean textual data suitable for vectorization and classification is attained

1.3.3 Feature Extraction

After preprocessing the text, the next important step in the text classification process is converting the cleaned text data into a numerical format that machine learning algorithms can work with. This step, is crucial for the performance and understanding of the classification models. In this project, the Term Frequency–Inverse Document Frequency (TF-IDF) method is chosen as the main technique for feature extraction.

TF-IDF (Term Frequency–Inverse Document Frequency) is a statistical technique utilized in natural language processing and information retrieval to assess the significance of a word within a document compared to a broader set of documents (Anon., 2025). TF-IDF consists of two parts:

1. **Term Frequency (TF):** This measures the frequency of a word in a document. A higher frequency indicates greater significance. When a term is frequently found in a document, it is probably relevant to the content of that document (Anon., 2025).

$$TF(t, d) = \frac{\text{Number of times term } t \text{ appears in document } d}{\text{Total number of terms in document } d}$$

Figure 3 Term Frequency (Anon., 2025)

2. **Inverse Document Frequency (IDF):** Lowers the importance of common words found in many documents and raises the importance of less common words. A term that shows up in fewer documents is usually more significant and specific (Anon., 2025).

$$IDF(t, D) = \log \frac{\text{Total number of documents in corpus } D}{\text{Number of documents containing term } t}$$

Figure 4 Inverse Document Frequency (Anon., 2025)

This balance enables TF-IDF to emphasize terms that are common in a particular document and unique throughout the text, making it a valuable tool for activities such as search ranking, text classification, and keyword extraction.

Reasons for Selecting TF-IDF:

The selection of TF-IDF was influenced by various task-related, educational, and methodological factors, especially regarding the classification of cognitive levels based on Bloom's Taxonomy.

1. Compatibility with Bloom's Taxonomy Framework

Bloom's Taxonomy is fundamentally driven by verbs, where each cognitive level is typically indicated by specific instructional verbs like define, explain, analyze, evaluate, and design. These verbs act as clear linguistic markers of the cognitive demand intended by a question. TF-IDF is particularly adept at identifying such distinguishing terms because:

- It emphasizes words that frequently appear within a particular cognitive category.
- It reduces the weight of generic terms that are found across different categories.

Consequently, TF-IDF effectively highlights action verbs specific to Bloom's framework, making it very appropriate for this classification task.

2. Effectiveness for Short Educational Texts

This project uses a dataset mainly made up of short to medium-length educational questions. TF-IDF has proven to be very effective in these situations because:

- Short texts typically depend on a limited number of key terms to express their meaning.
- Instructional verbs and phrases hold significant semantic weight.

TF-IDF effectively identifies these features without needing extensive training data.

1.3.4 Machine Learning Models for Cognitive Level Classification

1.3.4.1 Multinomial Naïve Bayes

Multinomial Naive Bayes is a type of the Naive Bayes algorithm that works best with discrete data and is commonly applied in text classification tasks. It represents the frequency of words as counts and assumes that each feature or word follows a multinomial distribution, its frequently used for tasks such as classifying documents based on word frequencies (Anon., 2025).

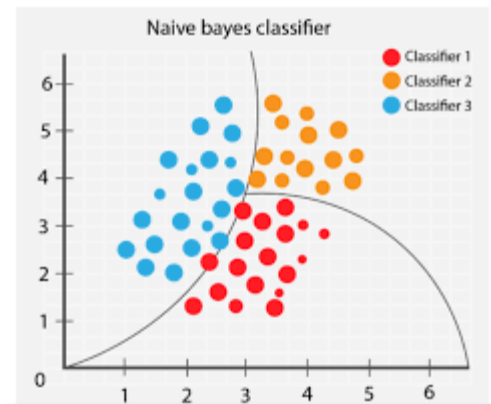


Figure 5 Naive Bayes Classifier (R, 2024)

To calculate the probability of a message belonging to a certain category Multinomial Naive Bayes uses the multinomial distribution:

$$P(X) = \frac{n!}{n_1!n_2!\dots n_m!} p_1^{n_1} p_2^{n_2} \dots p_m^{n_m}$$

Figure 6 Multinomial Distribution (Anon., 2025)

Where:

- n represents the total number of trials.
- n_i is the number of times outcome i occurs.
- p_i is the probability of outcome i .

To determine the likelihood of each word belonging to a specific category we apply a technique known as Maximum Likelihood Estimation (MLE).

$$\theta_{c,i} = \frac{\text{count}(w_i, c) + 1}{N + v}$$

Figure 7 Maximum Likelihood Estimation (Anon., 2025)

Where:

- $\text{count}(w_i, c)$ indicates how many times the word w_i appears in the documents belonging to class c .
- N is the total count of words in the documents of class c .
- v represents the size of the vocabulary.

The main benefits of Multinomial Naïve Bayes are:

- Quick computation and rapid training time
- Good baseline results for text-related tasks
- Fits well with term-frequency features
- Simple to implement and easy to understand

1.3.4.2 Logistic Regression

Logistic regression, similar to linear regression, is a kind of linear model that looks at the connection between predictor variables (independent variables) and an output variable (the response, target, or dependent variable). The main distinction is that linear regression is applied when the output is a continuous value like predicting a person's credit score. In contrast, logistic regression is utilized when the result is categorical, such as determining if a loan is approved or not. In logistic regression, the model forecasts the likelihood that a particular outcome will happen (Lee, n.d.).

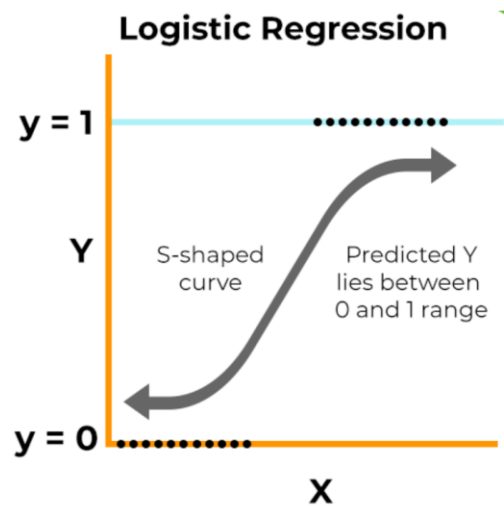


Figure 8 Logistic Regression (Kanade, 2022)

Logistic regression employs a logistic function known as the sigmoid function to translate predictions into probabilities, the sigmoid function is an S-shaped curve that transforms any real number into a value between 0 and 1 (Kanade, 2022).

$$y = \frac{e^{(b_0 + b_1 X)}}{1 + e^{(b_0 + b_1 X)}}$$

4

Figure 9 Sigmoid Function (Kanade, 2022)

The main benefits of Logistic Regression are:

- It efficiently scales with extensive feature spaces
- It generates probabilistic results for every class
- It enables direct examination of feature weights
- It functions effectively when class boundaries intersect

1.3.4.3 Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised learning algorithm used for both classification and regression tasks. It aims to identify the optimal boundary, called a hyperplane, that divides different classes within the data. This method is particularly effective for binary classification problems, such as distinguishing between spam and not spam or between cats and dogs. The main concept of the SVM algorithm is to locate the hyperplane that most effectively separates the two classes by maximizing the margin between them. This margin represents the distance from the hyperplane to the closest data points (support vectors) on either side.

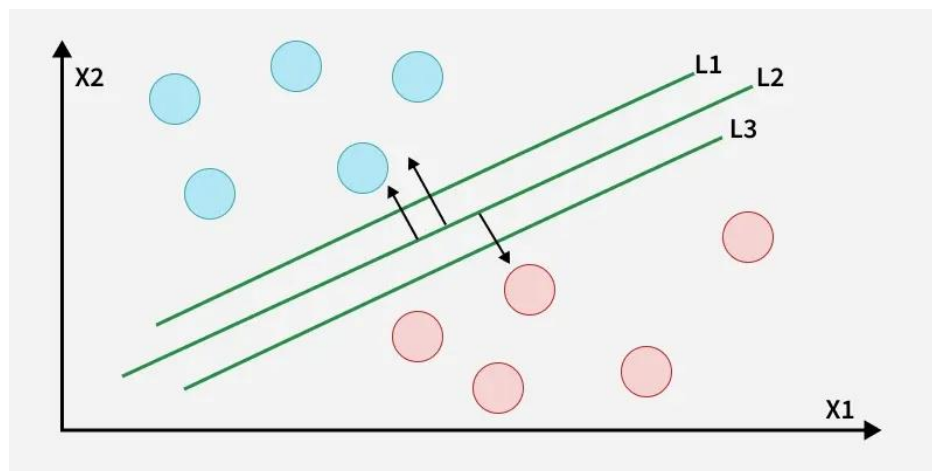


Figure 10 Support Vector Machine (Anon., 2025)

2. Problem Domain

2.1 Problem Domain: Cognitive Level Classification of Educational Questions

As digital education platforms grow quickly, a large amount of educational content like quizzes, assignments, exam questions, and teaching materials is produced every day. Properly organizing, analyzing, and assessing this content has become more crucial for improving teaching quality, boosting student engagement, and creating effective assessments.

One significant challenge in this area is understanding and identifying the cognitive complexity of educational questions, which directly affects learning outcomes and the quality of assessments. Educational questions aim not just to test factual knowledge but also to assess higher-order thinking skills such as reasoning, problem-solving, and critical evaluation. However, figuring out the cognitive level of a question by hand is labor-intensive and often subjective, relying on the educator's interpretation and experience. This issue emphasizes the need for automated and consistent methods for classifying cognitive levels.

Bloom's Taxonomy as a Cognitive Framework

Bloom's Taxonomy is a well-known educational framework that classifies learning goals based on the cognitive skills and abilities they need. It has been essential in designing curricula, developing assessments, and planning instruction for many years and is seen as one of the most impactful frameworks for curriculum development in the 21st century (Seyyed Mohammad Ali Soozandehfar, 2016).

The original taxonomy, created in 1956, emphasized a hierarchical structure of cognitive skills. In 1999, Dr. Lorin Anderson, a former student of Benjamin Bloom, and his team proposed an updated version of Bloom's Taxonomy that fixed some issues with the original framework. This new taxonomy offered a more dynamic and learner-focused approach by differentiating between 'knowing what' and 'knowing how' (Seyyed Mohammad Ali Soozandehfar, 2016).

The updated taxonomy has two parts:

The Knowledge Dimension, which shows what we learn ("knowing what"). This dimension is split into four types:

- Factual knowledge
- Conceptual knowledge
- Procedural knowledge
- Metacognitive knowledge

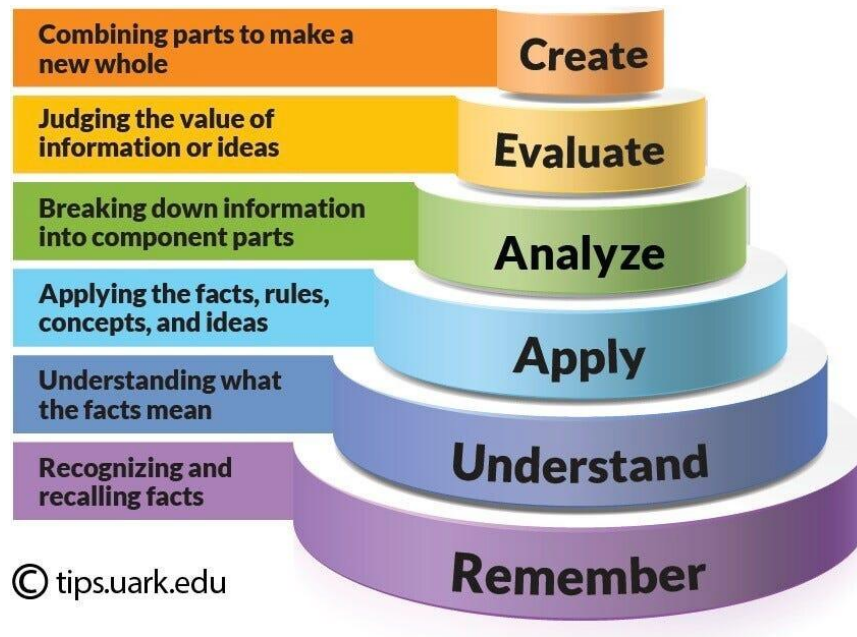
The Cognitive Process Dimension, which shows how we learn and think ("knowing how"). This dimension, like the original Bloom's Taxonomy, has six levels of cognitive skills, from simpler to more complex:

- Remember
- Understand
- Apply
- Analyze
- Evaluate
- Create

(Seyyed Mohammad Ali Soozandehfar, 2016)

Remembering is the ability to identify and recall important information from long-term memory. Understanding means making sense of instructional messages, which includes interpreting, summarizing, classifying, comparing, and explaining ideas. Applying is about using learned methods in both familiar and new situations. Higher levels of the taxonomy need more complex thinking. Analysis means breaking down information into parts and looking at how they relate. Evaluation is about making decisions based on set criteria and standards, while creation is about coming up with new ideas, products, or perspectives on concepts.

Correctly categorizing educational questions based on Bloom's Taxonomy is very important for today's education systems. It aids teachers in creating fair assessments, helps align learning goals with evaluation techniques, and allows adaptive learning platforms to customize content according to students' cognitive requirements.



(Anon., 2022)

2.2 Description and background research on the chosen dataset

The dataset for this project was thoughtfully assembled to aid in classifying educational questions based on Bloom's Taxonomy cognitive levels. Instead of using just one dataset, various publicly accessible academic datasets were combined to create a varied and representative collection of educational questions. This method strengthens the dataset's reliability, minimizes domain-specific bias, and enhances the general applicability of the trained classification models.

1. Mendeley Dataset: Computer Science Course Questions Organized by Bloom's Taxonomy

Source:

Dataset of Computer Science Course Queries from Students: Categorized and Scored According to Bloom's Taxonomy.

<https://data.mendeley.com/datasets/w5zt9n6vsc/2>

Data format: CSV

Dataset Features:

The main dataset for this study is called "Dataset of Computer Science Course Queries from Students: Categorized and Scored According to Bloom's Taxonomy". It was published on 8 January 2024 as Version 2 in the Mendeley Data repository. This dataset is available to the public under the Creative Commons Attribution 4.0 (CC BY 4.0) license and can be accessed via the DOI: 10.17632/w5zt9n6vsc.2.

This dataset was provided by Khandoker Ashik Uz Zaman, Yusuf Mahbubul Islam, Md Abu Sayed, and Ashraful Islam. It includes a well-organized collection of genuine educational queries created by students during the Summer 2023 semester at Independent University, Bangladesh (IUB).

Data Gathering Method:

The dataset was gathered through an online question-submission tool used in actual classroom settings. Students taking computer science classes utilized the tool to ask

questions during lectures and learning sessions. This method of collecting data in real-time made sure that the questions genuinely represented students' engagement, curiosity, and understanding of concepts throughout the learning experience. The dataset focuses on questions from two undergraduate computer science courses:

- Data Structures
- Introduction to Computers and Research

Along with pertinent academic questions, the system also recorded a number of irrelevant questions, which were kept in a separate file for research clarity and future filtering studies.

Data Composition:

The dataset includes three CSV files, each with a specific role:

- `Data_Structure.csv`
This file has questions about the Data Structures course, organized by topic and rated based on Bloom's Taxonomy.
- `Introduction_to_Computers_and_Research.csv`
This file contains questions from the Introduction to Computers and Research course, also organized and rated.
- `Irrelevant_Questions.csv`
This file holds questions from students that are not directly related to the course material.

Each relevant CSV file has these attributes:

- Topic: The course topic linked to the question
- Score: A numerical value indicating the Bloom's Taxonomy cognitive level
- Questions: The actual text of the question submitted by students.

For this project, only the relevant course-specific datasets (first two datasets) were utilized, while the irrelevant questions were left out to keep the focus on valuable educational content.

Structure:

Initially, the dataset had three columns:

Topic: The subject area of the question

Score: A numerical representation of Bloom's cognitive level

Questions: The text of the question

During the preprocessing stage, the Topic column was eliminated as it does not directly aid in the cognitive-level classification task and could introduce unintended topical bias. The numerical scores were aligned with standard Bloom's Taxonomy labels (Knowledge, Comprehension, Application, Analysis, Evaluation, Creation).

The two datasets on aggregation resulted in 3,104 rows.

2. Figshare Dataset: Examination Questions Annotated Using Bloom's Taxonomy

Source:

Exam Question Datasets

Figshare Repository

https://figshare.com/articles/dataset/Exam_Question_Datasets/22597957

Dataset posted: 26 April 2023

Authors: Mohammed Osman Gani, Anbuselvan Sangodiah

Data format: CSV

Dataset Features:

The Figshare dataset named "Exam Question Datasets" includes exam-style questions that are clearly categorized based on Bloom's Taxonomy Cognitive Domain. This dataset was developed to aid research in educational data mining, question classification, and automated assessment analysis. Consequently, the dataset offers excellent examples of the application of Bloom's Taxonomy in organized academic assessments.

Data Gathering Method:

The dataset was created by gathering and combining several question corpora annotated with Bloom's Taxonomy from recognized academic research. Initially, a dataset called collected_dataset.csv was put together and labeled by the authors. Additionally, three external datasets were added, sourced from previous studies that focused on classifying questions based on Bloom's Taxonomy.

- Bloom's Taxonomy–Based Classification for Item Bank Questions Using Support Vector Machines, published in Modern Advances in Intelligent Systems and Tools. Yahya, A. A., Toukal, Z., and Osman, A. (2012).

This research examined the application of Support Vector Machines (SVM) for the automatic classification of exam questions according to Bloom's cognitive levels.

- Taxonomy-based features in question classification using Support Vector Machine, published in the Journal of Theoretical and Applied Information Technology.

Sangodiah, A., Ahmad, R., and Ahmad, W. F. W. (2017)

This study looked into how effective taxonomy-driven linguistic features are for classifying cognitive levels.

- Question classification based on Bloom's taxonomy cognitive domain using modified TF-IDF and Word2Vec, published in PLoS ONE.

Mohammed, M., and Omar, N. (2020)

This research showed that merging traditional TF-IDF representations with semantic embeddings enhances classification performance.

Besides these three studies, the dataset also contains one more Bloom's Taxonomy–annotated question corpus obtained from related academic research, as mentioned by the authors. Ultimately, all five datasets (the authors' manually labeled dataset and the four derived from literature) were combined into a single file named `combined_dataset.csv`, creating a thorough Bloom's Taxonomy–annotated question corpus.

Structure:

Columns:

1. Questions: The natural language text of examination questions.
2. BT Level: The Bloom's Taxonomy cognitive category assigned to each question.

Rows: 2522

Domain coverage: Multidisciplinary (education, economics, computing, science, etc.)

3. Kaggle Dataset: Bloom's Taxonomy–Labeled Examination Questions

Source:

Bloom's Taxonomy Dataset

Kaggle Repository

<https://www.kaggle.com/datasets/vijaydevane/blooms-taxonomy-dataset/data>

Data format: CSV

Dataset Features:

The third dataset utilized in this research is the Bloom's Taxonomy Dataset sourced from the Kaggle platform. This dataset consists of a structured set of educational questions organized by the six cognitive levels of Bloom's Taxonomy: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating.

The dataset is intended to aid in the creation of educational tools, smart assessment systems, and machine learning models that seek to automatically evaluate and categorize the cognitive complexity of learning resources.

Data Gathering Method:

The dataset was created by collecting questions from AICTE (All India Council for Technical Education) exam paper samples, which are publicly accessible academic assessment resources. The original exam questions were obtained from official and educational websites that provide AICTE model question papers. Once collected, the questions were reworded and organized into a consistent textual format suitable for automated classification. Each question was then categorized according to the relevant Bloom's Taxonomy level based on:

- The action verbs present in the question
- The cognitive demand needed to answer the question
- The learning outcome being evaluated

This approach guarantees that the dataset is closely aligned with actual examination and curriculum design practices.

Structure:

Columns:

1. Questions: The natural language text of examination questions.
2. BT Level: The Bloom's Taxonomy cognitive category assigned to each question (BT1–BT6).

Rows: 8767

In the preprocessing stage, the BT1–BT6 labels were assigned to their related categories (Knowledge, Comprehension, Application, Analysis, Evaluation, Creation) to maintain consistency with other datasets utilized in this research.

2.3 Brief discussion of the societal or business relevance of the chosen classification task.

The automatic classification of educational questions based on Bloom's Taxonomy is very important for society, education, and business, especially in the fast-growing digital education landscape. As online learning platforms, smart tutoring systems, and large assessment tools expand, the amount of educational content produced has surged dramatically. Analyzing and categorizing this content by hand is time-consuming and subjective, making automated solutions more essential.

From a societal viewpoint, classifying questions using Bloom's Taxonomy helps enhance the quality and fairness of education. Educators need to create assessments that measure students at various cognitive levels instead of just focusing on memorization. However, achieving a balanced range of cognitive difficulty in tests and assignments is tough when done manually.

An automated classification system can help educators by:

- Checking if assessments match the intended learning outcomes
- Ensuring a good mix of lower-order and higher-order cognitive questions

These systems can also aid students by fostering deeper learning through exposure to questions that require analysis, evaluation, and problem-solving instead of just recall.

From a business standpoint, the classification based on Bloom's Taxonomy is vital for creating EdTech products and AI-based learning platforms. Companies in educational technology are progressively depending on smart systems to tailor learning experiences, create assessments, and assess student performance.

Important business uses include:

- Intelligent tutoring systems that modify question difficulty according to student performance
- Automated question generation tools that guarantee cognitive diversity in assessments

- Learning management systems (LMS) that evaluate and structure instructional content
- AI-driven curriculum design tools utilized by educational institutions and training organizations

3. SOLUTION

3.3 Diagrammatic Representation

3.3.1 Flowchart for overall system

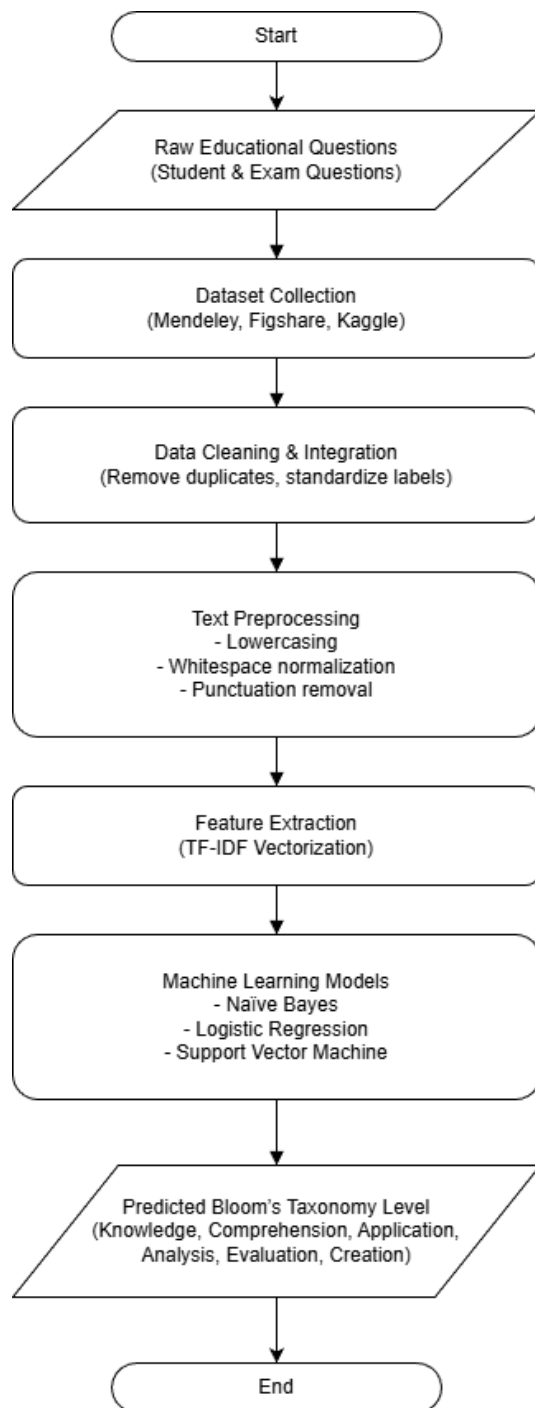


Figure 11 Flowchart for overall system

3.3.2 Flowchart for TF-IDF

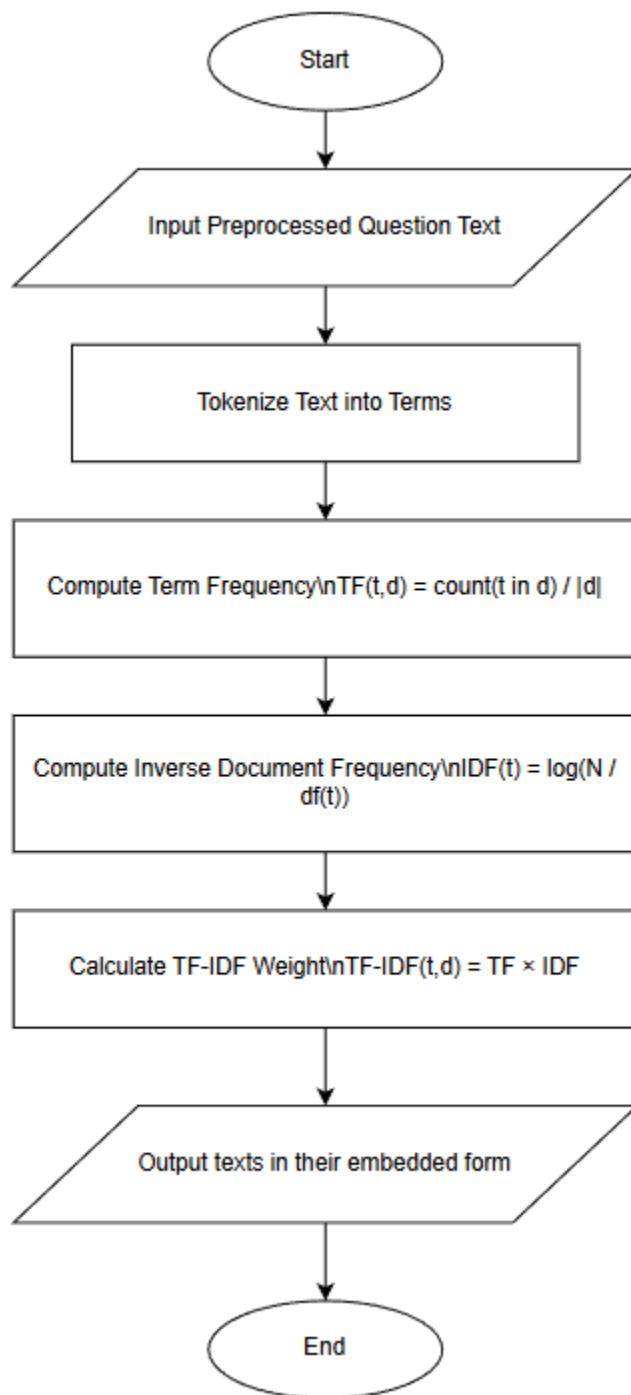


Figure 12 Flowchart for TF-IDF

3.3.3 Flowchart for Naïve Bayes

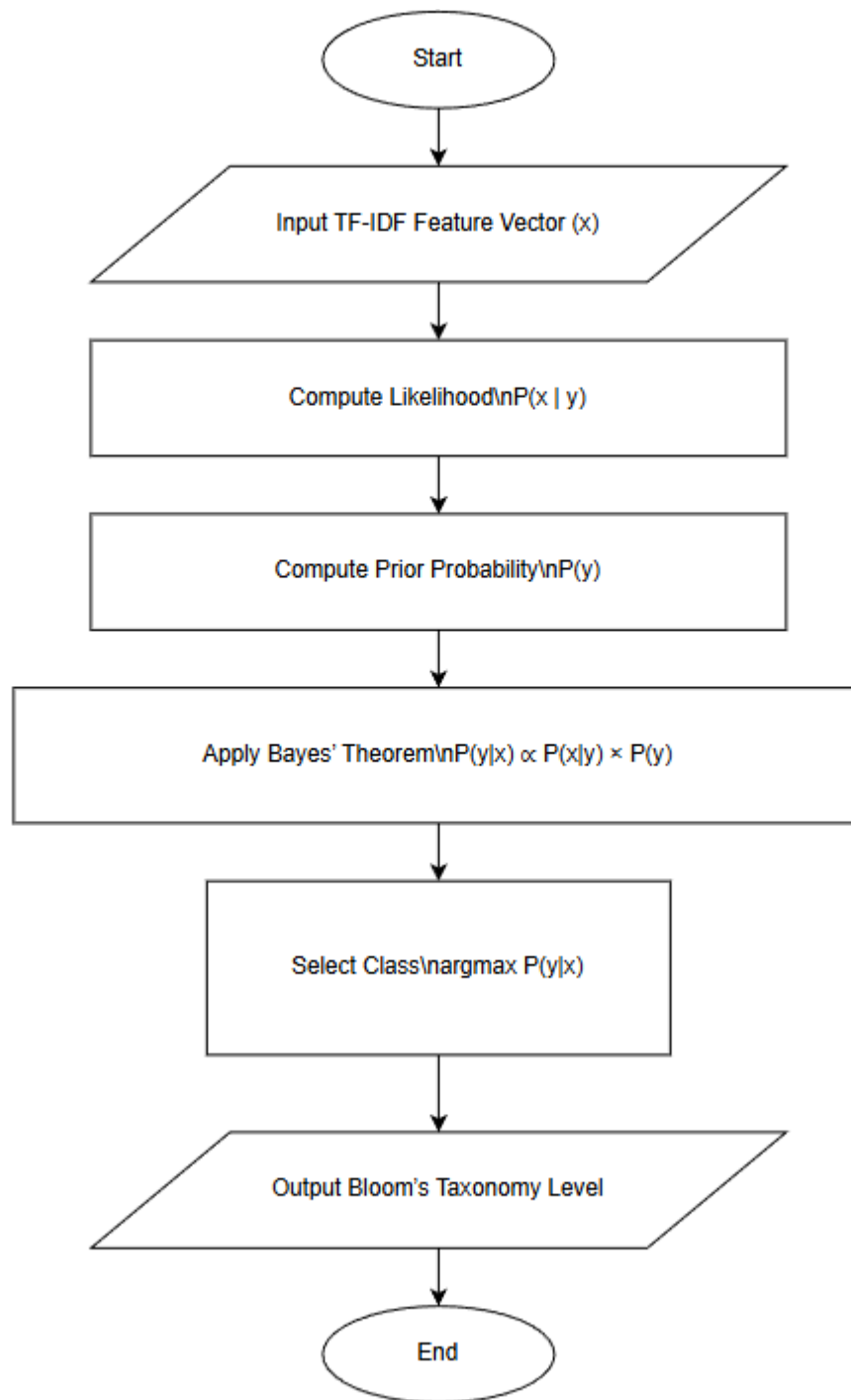


Figure 13 Flowchart for Naïve Bayes

3.3.4 Flowchart for Logistic Regression

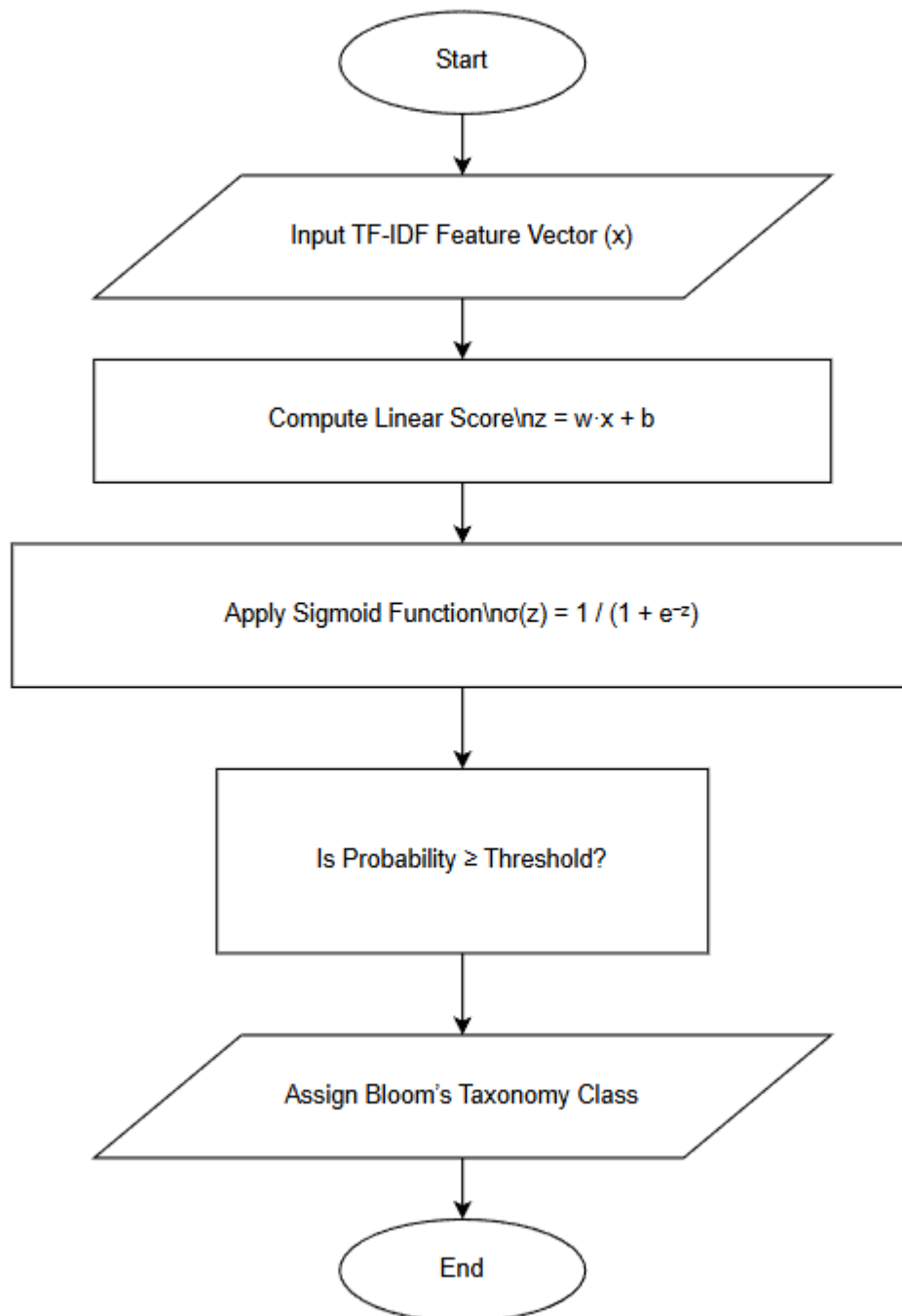


Figure 14 Flowchart for Logistic Regression

3.3.5 Flowchart for SVM

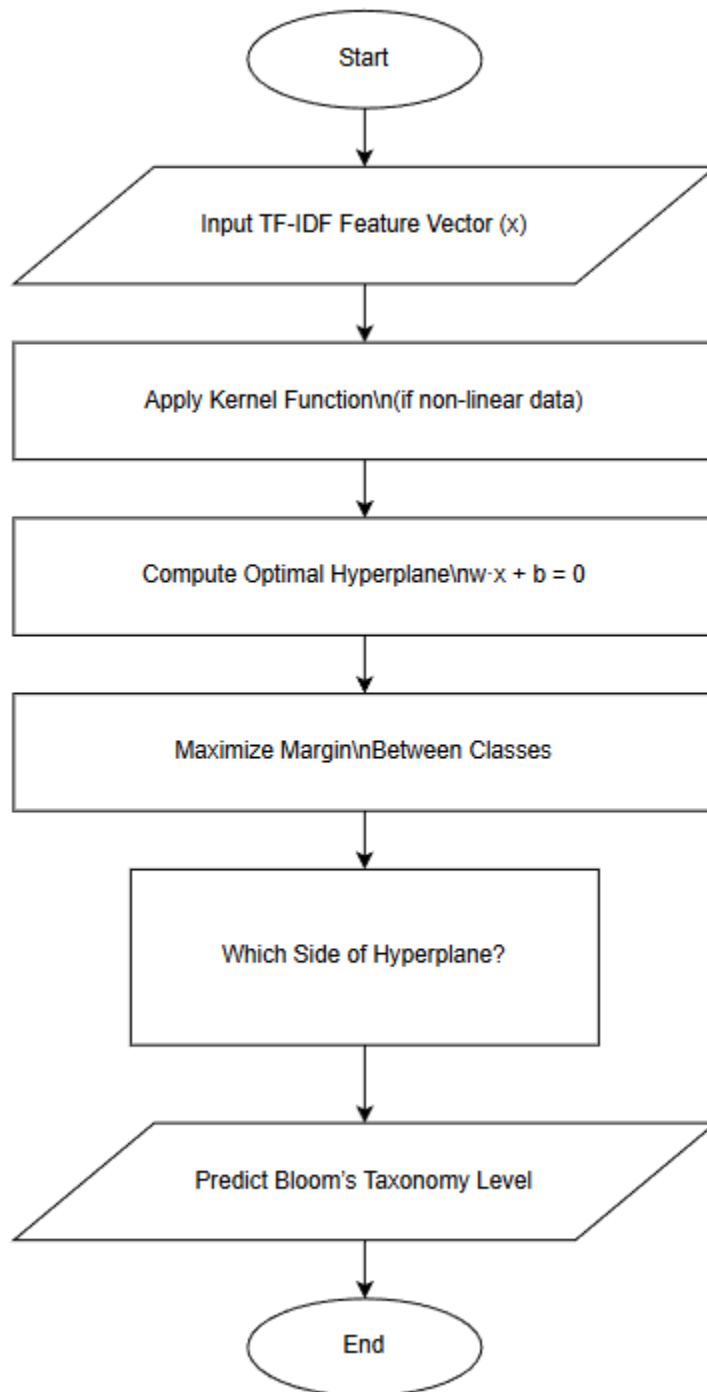


Figure 15 Flowchart for SVM

3.3.6 State Diagram

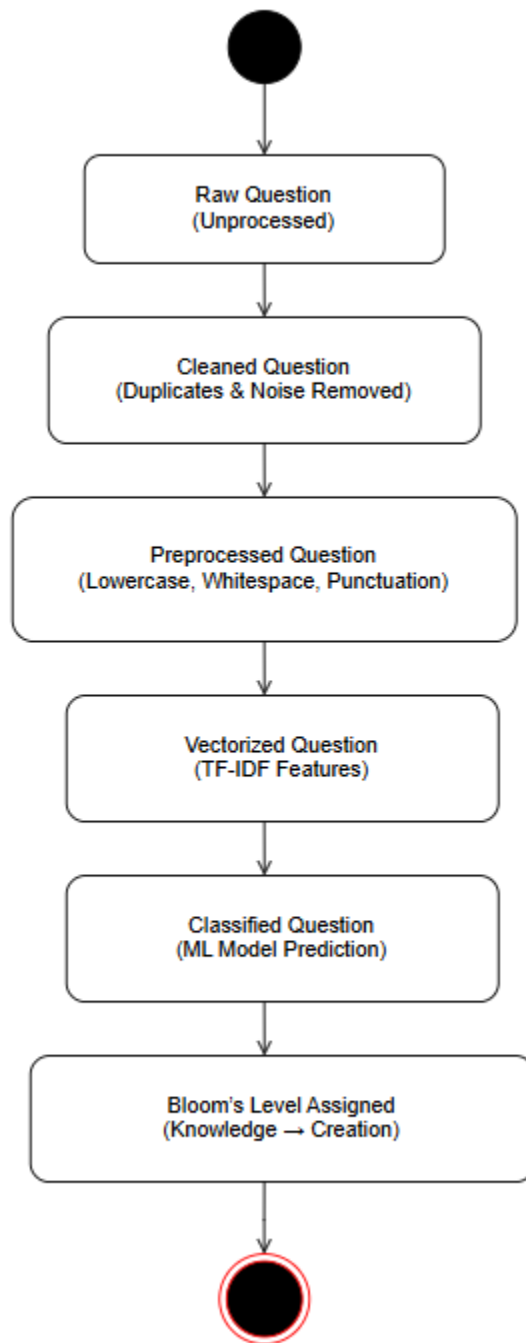


Figure 16 State Diagram

3.3.7 System Architecture Diagram

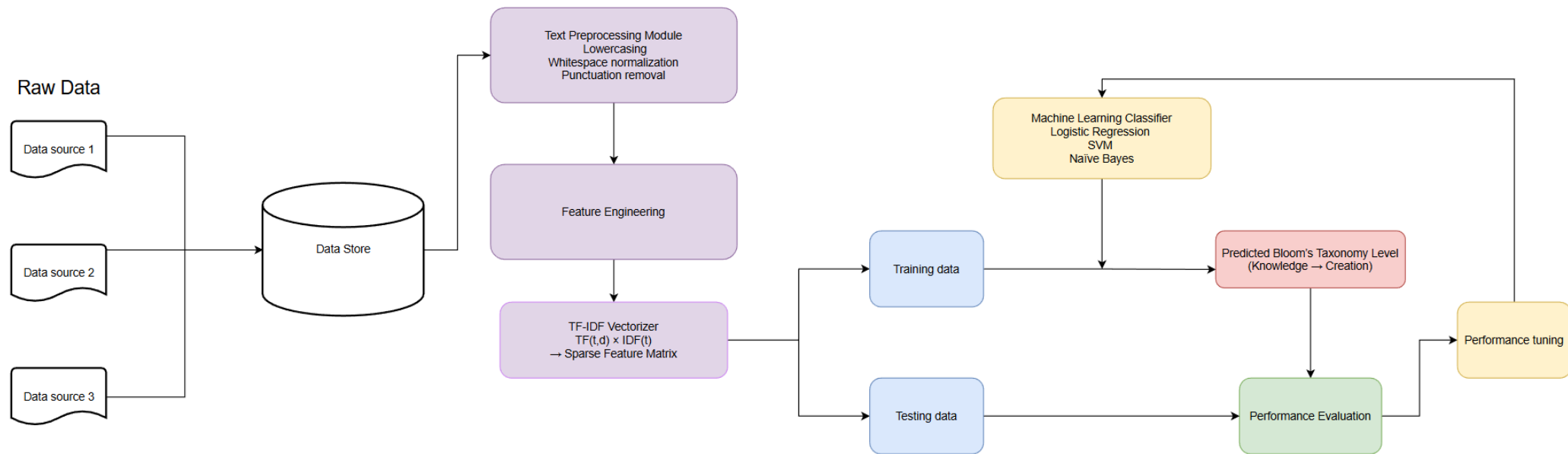


Figure 17 System Architecture Diagram

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